Multi Party computation

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Notes from Shoup

Keywords:

1 Garbled circuit

Garbled Circuit	Beaver's
Boolean circuit	Arithmetic circuit
Constant rounds of communication needed	#Rounds depend on #multiply gates
3-party protocol	2.5 party protocol
In preprocessing as well as lineexecution 3rd party is involved	Processor notinvolved during line execution of protocol

2 Yao's 2 party GC

Garble (f)

- Circuit garbling algorithm
- (f) : boolean circuit
- e: input encoding data
- d: input decoding data
- ($\mathfrak{F}$, e, d) $\leftarrow$ Garble(f)

$\underline{F_{encode}}$

- $\mathfrak{X} \leftarrow$ Encode(e,x), where x is a vector of bits (input of all parties)

($\underline{F}_{encode}$)

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